Jakkula Adishesh Balaji

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.64
FIITJEE Junior College	Class XII (Telangana State Board)	2024	96.30%
FIITJEE World School	Class X (Telangana State Board)	2022	9.50%

SCHOLASTIC ACHIEVEMENTS

- One of the **seven finalists** at **Convolve 4.0 Pan IIT Hackathon** (Generative AI Track) out of **10000+ Teams**
- **Third Place** at **Image Restoration Hackathon** Held by **KLA** at IITH

POSITIONS OF RESPONSIBILITY

- **AI/ML Core Member**, **Epoch** - *(FY 25-26)*
- **Dev Core Member**, **Lambda** - *(FY 25-26)*
- **AI/ML Core Member**, **Tinkerer's Lab** - (*FY 25-26)*

SKILLS

- **Programming and Scripting Languages**: Python, C++, C, MySQL, HTML, CSS, JavaScript
- **AI & ML**: Pytorch, Scikit-Learn, Numpy, Pandas, HF Transformers
- **Web & Tools**: Git, Github, Linux, React, Node.js, Express, FastAPI

RELEVANT COURSES

- **AI/ML** : Foundations of ML, Deep Learning, AI For Humanity, Programming for AI
- **Computer Science**: Data Structures and Applications, Algorithms, DBMS I, Discrete Maths
- **Mathematics**: Probability, Calculus I & II, Matrix Theory, Concentration Inequalities

PROJECTS

NoteSage | Multi-Document RAG System (Github)

- Developed a **Hybrid RAG pipeline** in Python for academic PDFs, combining **ChromaDB** semantic search and **BM25 lexical retrieval** using **Reciprocal Rank Fusion (RRF)**.
- Enhanced retrieval quality with **cross-encoder reranking**, **query rewriting**, and **OCR-based PDF ingestion** for accurate multi-document question answering.
- Deployed an interactive streamlit UI and built an automated **LLM-as-a-Judge** evaluation framework using **OpenRouter (gpt-oss-120b)** to benchmark responses on faithfulness, relevance, and completeness

Evaluating xAI Robustness via Adversaries (Github)

- Engineered an adversarial pipeline combining **OOD detection, MLP gradient regularization,** and **fairness-constrained Logistic Regression** to manipulate **LIME/SHAP** explanations.
- Designed **OOD-based scaffolding** and **gradient-guided attacks** that preserved **Demographic Parity** on real test data while misleading feature attributions toward sensitive attributes.
- Achieved **80% accuracy** and **0.73 F1-score**, and explored **CAD-Defend** as a mitigation strategy.

Image Denoising Model (Github)

- Implemented a high-performance image restoration model as part of the **KLA Hackathon** held at IITH using **NAFNet**

and **Discrete Wavelet Transform(DWT)** to effectively isolate noise from structural details in the frequency domain.
- Optimized for deployment with a **5.5M parameter footprint**, achieving **52ms average inference latency** through advanced techniques like mixed-precision inference, tile-based processing and model compilation
- Attained a validation PSNR of **29.12 dB** by integrating a **hybrid Charbonnier-FFT loss function**, ensuring the precise preservation of high-contrast boundaries and textures while suppressing noise.

Sentiment Analysis on IMDb Reviews (Github)

- Built and compared 16 PyTorch sentiment classification models (RNN/LSTM/BiRNN/BiLSTM) along with Bahdanau, Luong Dot, Luong Concat, Luong General attention mechanisms on the Stanford IMDb dataset
- Fine-tuned a Luong General Attention LSTM using 100 dimensional GloVe embeddings, achieving **89.85% test accuracy** and **0.90 F1 score**, with real-time attention weight visualization for model interpretability
- Deployed the application as a full-stack system using **React (Vite), FastAPI**, **ONNX Runtime**, **Render**, and **Vercel**, enabling low-latency inference and interactive sentiment analysis.

EXTRACURRICULAR

- **1st Rank**, **NSF(North South Foundation)** Junior Spelling Bee - **National (2015-16)**, **Regional (2014-15)**
- Rated **2036** in **Rapid Format** at **chess.com** placing me in the top **0.2%** of players worldwide
- Participated and qualified several **Maths and Science olympiads(SOF)** at School Level

RESEARCH EXPERIENCE

Robust Deepfake Detection | (Guided by Prof. Sumohana Channappayya) (Github)

- Implemented a **multi-modal pipeline** combining **Vision Transformers (ViT)** for RGB spatial features, CNNs for **Haar DWT frequency subbands**, and LSTMs for sequential temporal tracking
- Deployed the training pipeline on the **PARAM Seva supercomputing cluster** using **SLURM.**
- Visualized **hidden-layer activations** using **t-SNE** to analyze feature separability to diagnose model performance